

Phase 2: Specialisation Tracks

Track 1: SAS to Azure - Data Scientist Track

Overview

Format:

- 40 hours over 6 weeks (part-time continuous learning)
- 1 day per week commitment (½ day live + ½ day self-paced)
- Learning format: 50% synchronous / 50% asynchronous learning

Objective: Transition from SAS to Azure-based analytics and machine learning workflows (leveraging Python within Azure Databricks).

Week 1: Python Fundamentals in Azure Databricks (7 hours)

Live Session (3.5 hours)

- Introduction to Python syntax within Azure Databricks notebooks
- Variables and data types in a distributed notebook environment
- Basic scripting for Azure-connected datasets
- SAS DATA step logic → Python implementation in Databricks

Self-Paced (3.5 hours)

- Exercises executed directly in Allianz Databricks workspace
- Working with Azure-connected sample datasets

Week 2: Advanced Python & Object-Oriented Programming (7 hours)

Live Session (3.5 hours)

- Classes and objects: instantiation, methods, attributes
- Modules and packages: importing, using libraries
- Error handling: try/except blocks
- File I/O: reading and writing CSV, JSON

Self-Paced (3.5 hours)

- Exercises: Build a data processing class similar to SAS macros
- Quiz: OOP concepts and modules
- Read documentation on Python standard library

Week 3: Pandas for Data Manipulation - Part 1 (7 hours)**Live Session (3.5 hours)**

- Introduction to Pandas: Series and DataFrames
- Loading data from CSV, Excel, Azure SQL
- Initial data exploration: head(), info(), describe()
- Selecting, filtering, and sorting data
- SAS comparison: DATA step → Pandas DataFrame operations

Self-Paced (3.5 hours)

- Exercises: Load Allianz dataset and perform basic filtering
- Quiz: Pandas basics and data selection
- Compare Pandas code with equivalent SAS DATA step

Week 4: Pandas for Data Manipulation - Part 2 (7 hours)**Live Session (3.5 hours)**

- Data cleaning: handling missing values, duplicates
- Data transformation: apply(), map(), lambda functions
- GroupBy operations and aggregations
- Merging and joining DataFrames
- SAS comparison: PROC SQL → Pandas merge/join

Self-Paced (3.5 hours)

- Exercises: Clean insurance claims data, perform aggregations
- Quiz: Data transformation and GroupBy
- Project checkpoint: Define capstone project scope

Week 5: Azure Databricks, MLflow & AML (7 hours)**Live Session (3.5 hours)**

- Azure Databricks workspace overview

- Running Python notebooks in Databricks
- Connecting to Azure Data Lake Storage and Azure SQL
- Introduction to MLflow for experiment tracking
- Logging parameters, metrics, and models
- Introduction to Azure Machine Learning service
- Model registration and deployment concepts in Azure ML
- Managing the ML lifecycle within the Azure ecosystem

Self-Paced (3.5 hours)

- Exercises: Run ML experiments in Databricks with MLflow tracking
- Explore Azure ML Studio interface
- Quiz: Databricks, MLflow and Azure ML concepts
- Continue capstone project in Azure environment

Week 6: Capstone Project & Final Review (6 hours)

Live Session (3 hours)

- Project mentoring session: review progress and troubleshoot
- Best practices for production ML in Azure
- Code review and optimization tips
- Final Q&A and course wrap-up

Self-Paced (3 hours)

- Finalize capstone project
- Prepare project presentation
- Submit final deliverables

Capstone Project - Data Scientist Track

Migrate a SAS-based analytical workflow to the Azure data platform using Databricks and Azure ML services. Project includes:

- Rebuild SAS data preparation workflows within Azure Databricks
- Build a machine learning model (e.g., insurance risk prediction)
- Track experiments with MLflow & connect to Azure data sources (SQL, ADLS)
- Document code and present findings to peers